

Keshav Bimbraw

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EDUCATION

Worcester Polytechnic Institute, Worcester, MA

Ph. D. Candidate in Robotics Engineering (Medical FUSION Lab)

Aug '20 – May '25

Concentration – Medical Robotics, Deep Learning, Ultrasound Image Processing, & Human-Machine Interfacing

4.00/4.00 GPA

Georgia Institute of Technology, Atlanta, GA

M. S. with Robotics Focus (Music Technology, Computer Software & Media Applications)

Aug '17 – May '19

Concentration – Human Augmentation, Medical Robotics & Robotic Musicianship

3.56/4.00 GPA

Thapar University, Patiala, India

B. E. in Mechatronics Engineering (Research Intern - IIT Delhi, Jan – Jul '16)

July '13 – June '17

Concentration – Robotics & Mechatronics

4.00/4.00 GPA (8.34 CGPA)

EXPERIENCE

OnPoint Surgical

Concord, MA

Machine Learning Research and Development Engineer

Oct '24 – now

- Leading the AI efforts at OnPoint Surgical working as a machine learning software engineer in the surgical robotics space focusing on classification, segmentation and registration problems for 2D and 3D medical imaging modalities such as MRI, CT and X-Ray.
- Applied transfer learning and model quantization to deploy a MobileNetV2-based deep learning model for real-time AP/LA X-ray classification on edge devices with >98% accuracy and 4× size reduction. [Paper](#).

Mitsubishi Electric Research Laboratories

Cambridge, MA

Research Intern (Connected Interfaces Group, Mentor: Toshiaki Koike-Akino)

May '23 – Aug '23; June '24 – Aug '24

- Worked on exciting research directions focusing on Generative AI and Vision Language Models (VLMs) like GPT-4o. [Paper](#).
- Designed pipelines for multi-modal biosignal data acquisition and human-machine interaction in AR/VR/XR applications.
- Incorporated adversarial regularization to bolster hand gesture classification performance in the context of domain adaptation.
- Leveraged randomized channel ablation to optimize the performance of multi-modal biosignal modalities ([IEEE EMBC 2024](#)).

Worcester Polytechnic Institute (WPI)

Worcester, MA

Research Assistant (Medical FUSION Lab)

March '20 – now

- Developed a novel ultrasound-based method to simultaneously estimate manipulation skill type and grasp forces from forearm muscle activity, enabling real-time, high-resolution dexterity assessment for HMI applications.
- Funded by Amazon GBTI grant for the 2023-2024 academic year to use ultrasound to estimate finger forces to improve a haptic feedback system. Demonstrated ultrasound based continuous force estimation with an error of 5% (IEEE IUS '23). [Paper](#). [Video](#).
- Developed a reflector-based ultrasound system for gesture classification, utilizing CNNs and ViTs (SPIE MI '24). [Paper](#). [Journal](#).
- Developed a data-driven hand gesture recognition pipeline for VR interfacing (Meta Quest Pro) using ultrasound data and CNNs, achieving online accuracy of 92% for 4 gestures, and a low pipeline latency of 59.48 ms. [Paper](#). [Video](#).
- Developed a deep learning based pipeline to predict finger angles & hand configurations from forearm ultrasound images. Achieved an error of 7.35° for finger joint angle prediction (IEEE ICRA '22). [Paper](#). [Video](#). Extended version published in [IEEE T-MRB](#).
- Conceptualized a novel augmented reality based lung ultrasound scanning guidance system (MICCAI ASMUS '20). [Paper](#). [Video](#).
- Established the real-time ultrasound data acquisition software framework for a tele-operative low-cost lung ultrasound robot. [Paper](#).

Nokia Bell Labs

New Providence, NJ

Augmented Human Sensing Co-Op (Data and Devices Group, AI Research Lab)

June '22 – Dec '22

- Designed low-latency biosignal based pipelines (sEMG, IMU) to estimate bodily motion for Human-Machine Interaction. [Video](#).
- Submitted and presented a paper ([IEEE SII 2023](#)) and multiple invention reports. Offered a Co-Op extension, which was accepted.
- Developed demonstrations for biosignal-based remote multi-sensor real-time shared-autonomy industrial robotic applications.
- Investigated multi-channel sEMG based hand movement classification for robotic control using deep learning (MLP, CNN).
- Utilized transfer learning for improving performance of sEMG based hand configuration classification from 91% to 96%.

Agile Resources Inc.

Alpharetta/Peachtree City, GA

Audio DSP Engineer (Panasonic Automotive Systems of America)

Oct '19 – March '20

- Developed Qualcomm's Hexagon SDK for audio applications in Eclipse on Android framework development.

Bose Corporation

Stow, MA

Active Noise Control Engineering Intern (Automotive Systems Division)

May '19 – Oct '19

- Designed a sound synthesis framework for electric vehicles and implemented processor-optimized signal processing algorithms.

- Devised a physical car sound simulation system using Logitech pedals to evaluate models and reduce testing time.

Georgia Institute of Technology (Georgia Tech)

Atlanta, GA

Research Assistant (Robotic Musicianship Lab)

Aug '17 – May '19

- Led the ML based ultrasound gesture classification [project](#) from Aug '18 – May '19. Robot and code featured in [The Age of A.I.](#)
- Investigated a combination of Ultrasound & EMG data using supervised learning algorithms to enhance assistive robot control.
- Created an ultrasound in the loop tendon-based wearable exoskeleton for upper extremity rehabilitation of stroke survivors. [Link](#).
- Taught Computer-Aided Design (CAD), Fusion 360, 3D printing, and Mechatronics for Project Studio Course. [Link](#), [Link](#).

IIT Delhi

New Delhi, India

Research Intern (Autonomous Robotics Lab)

Jan '16 – Aug '16

- Conceptualized and programmed the system framework for two KUKA KR-5 robotic arms to collaboratively play a guitar. [Link](#).
- Improved performance of a 6 DOF motion platform at Simulator Development Division, Secunderabad, India. [Link](#).
- Developed a teach pendant and a control module to control virtual robots in RoboAnalyzer software. [Link](#).

SKILLS

Software Skills

Python, C++, C#, TensorFlow, PyTorch, ROS, Linux, NumPy, MATLAB, OpenCV

3D Design/3D printing

SolidWorks, Autodesk Inventor, Autodesk Fusion 360, Unity, TinkerCAD, Cura

Electrical/Electronic skills

Arduino, Raspberry Pi, Motors, Actuators, Hardware & software interfacing

Programming Frameworks

Shell Scripting, Git, Visual Studio Code, PyCharm, Version Control, AWS Cloud

RECENT ACADEMIC PROJECTS

Analyzing the performance of a cart pole system for different parameters for RL (WPI, Artificial Intelligence) Spring '23

- Analyzed an RL framework in OpenAI Gym to evaluate cart-pole stability under varying parameters using DQN & Q-learning.

Comparing low & high-dimensional forearm ultrasound-based hand state classification (WPI, Machine Learning) Spring '22

- Investigated the feasibility of using image compression techniques for extracting the latent structure of ultrasound images. [Link](#).

LEADERSHIP

- Mentored high school & undergraduate students towards ongoing projects through MLSC, MQP & REU (WPI), & VIP (GT).

SELECTED HONORS & AWARDS (Full list: [CV](#))

- Won PhD Backlin Scholarship to cover tuition and health insurance through WPI. Sept. '24
- Won IEEE EMBC NextGen Scholar Award. May '24
- Funded through Amazon Robotics GBTI Award for the 2023-2024 academic year (Tuition support & yearly award) August '23
- Selected for the 2021B cohort of Google Research's CSRMP. The first person at WPI to have ever been selected. Sept. '21
- Awarded PhD positions at WPI & Georgia Tech. Accepted the WPI offer. (Tuition support & \$31824 yearly award). March '20
- Scholarship to pursue master's at Georgia Tech (Tuition support & \$14100 yearly award). August '17
- Awarded merit scholarships and grants totaling INR 336,000 at Thapar University. June '17

SELECTED FIRST-AUTHOR PUBLICATIONS (Full list: [CV](#))

- [1] **Bimbraw, K.**, Steines, D. (2025, May). Transfer Learning and Quantization for Efficient AP vs. LA X-Ray View Classification on an Edge Device. In *MIDL*. [Paper](#).
- [2] **Bimbraw, K.**, Wang, Y., Liu, J., & Koike-Akino, T. (2024, July). GPT Sonography: Hand Gesture Decoding from Forearm Ultrasound Images via VLM. In *NeurIPS Advancements in Medical Foundation Models Workshop*. [Paper](#). Journal extension accepted to IEEE Access (27% acceptance rate).
- [3] **Bimbraw, K.**, Zhang, H. K., & Islam, B. (2024, September). Forearm Ultrasound based Gesture Recognition on Edge. In *IEEE-EMBS BSN 2024*. [Paper](#).
- [4] **Bimbraw, K.**, Liu, J., Wang, Y., & Koike-Akino, T. (2024, July). Random Channel Ablation for Robust Hand Gesture Classification with Multimodal Biosignals. In *2024 IEEE Engineering in Medicine and Biology Conference (EMBC)* (pp. 1-6). IEEE. [Paper](#).
- [5] **Bimbraw, K.**, & Zhang, H. K. (2024, April). Mirror-based Ultrasound System for Exploring Hand Gesture Classification through Convolutional Neural Network and Vision Transformer. In *Medical Imaging 2024* (Vol. 12932, pp. 218-222). SPIE. [Paper](#). Extension in *IEEE Sensors Journal* (25% acceptance rate).
- [6] **Bimbraw, K.**, Rothenberg, J., & Zhang, H. K. (2023, October). Leveraging Ultrasound Sensing for Virtual Object Manipulation in Immersive Environments. In *IEEE-EMBS International Conference on Body Sensor Networks 2023*. [Paper](#). [Video](#).
- [7] **Bimbraw, K.**, & Zhang, H. K. (2023, September). Estimating Force Exerted by the Fingers Based on Forearm Ultrasound. In *IEEE IUS*. [Paper](#). [Video](#).
- [8] **Bimbraw, K.**, & Zheng, M. (2023, January). Towards The Development of a Low-Latency, Biosignal-Controlled Human-Machine Interaction System. In *2023 IEEE/SICE International Symposium on System Integration (SII)* (pp. 1-7). IEEE. [Paper](#). [Video](#).
- [9] **Bimbraw, K.**, Nycz, C. J., Schueler, M. J., Zhang, Z., & Zhang, H. K. (2022, May). Prediction of MCP joint angles & Classification of Hand configurations based on Ultrasound Imaging of the Forearm. In: *IEEE ICRA* (pp. 91-97). IEEE. [Paper](#). [Video](#). [Journal](#): *IEEE Transactions on Medical Robotics & Bionics*.
- [10] **Bimbraw, K.**, Ma, X., Zhang, Z., Zhang, H. (2020). Augmented Reality-Based Lung Ultrasound Scanning Guidance. In: *ASMUS 2020*. [Paper Link](#). [Video](#).